

Chapter 10: Asymmetric Information and Agency

(a) Asymmetric Information

Asymmetric information encompasses (various) situations where buyers and sellers have different levels of information. Basic microeconomic theory usually includes an assumption that the market being analyzed exhibits perfect information. Under conditions of perfect information, all consumers and producers have complete information on all prices, as well as the quality of any good or service available in the market. Although information is never perfect in the real world, perfect information serves as a useful starting point.

(b) The LEMON PRINCIPLE:

The market for lemons is a 1970 paper, by the economist Akerlof which examines how the quality of goods traded in a market can degrade in the presence of information asymmetry between buyers and sellers. In American slang, a lemon is a car that is found to be defective only after it has been bought.

Suppose buyers can not distinguish between a high quality car (a peach) and a lemon. Then they are only willing to pay a fixed price for a car that averages the value of a peach and lemon together. But sellers know whether they hold a

peach or a lemon. Given the fixed price at which buyers will buy, sellers will sell only when they hold "lemons" and they will leave the market when they hold peaches. Eve

To illustrate the nature of the used-car example, suppose that nine used cars are to be sold that vary in quality from 0, meaning a lemon, to a high of 2, meaning a mint condition used car, a "cream puff". In fact, suppose that the nine cars have respectively quality levels (Q) given by the cardinally measured index values of $0, \frac{1}{4}, \frac{1}{2}, \frac{3}{4}, 1, 1\frac{1}{4}, 1\frac{1}{2}, 1\frac{3}{4}$ and 2. Under a cardinal index, a car with a value of 1 has twice the quality of a car with an index of $\frac{1}{2}$. The distribution of these cars is shown in the following figure, where the horizontal axis shows the quality level and the vertical axis shows the uniform probability.

Suppose further that the owner of a car knows its quality level exactly but that the potential buyers

uniform probability
↑
→ quality level

know only the distribution of quality. It is known that the owners have a reserve value on their cars, so that reserve value to the seller = $\$1,000 \times Q$. That is, the owners would sell their cars only if they could get at least $\$2,000$ for every unit of quality that the car has. On the other hand, the nonowners are more eager for used cars and value them so that the value to nonowners = $\$1,500 \times Q$. To make this experiment a complete market, suppose that an auctioneer is hired to call out market prices; sales take place when the auctioneer finds a price that successfully equates quantity demanded with quantity supplied.

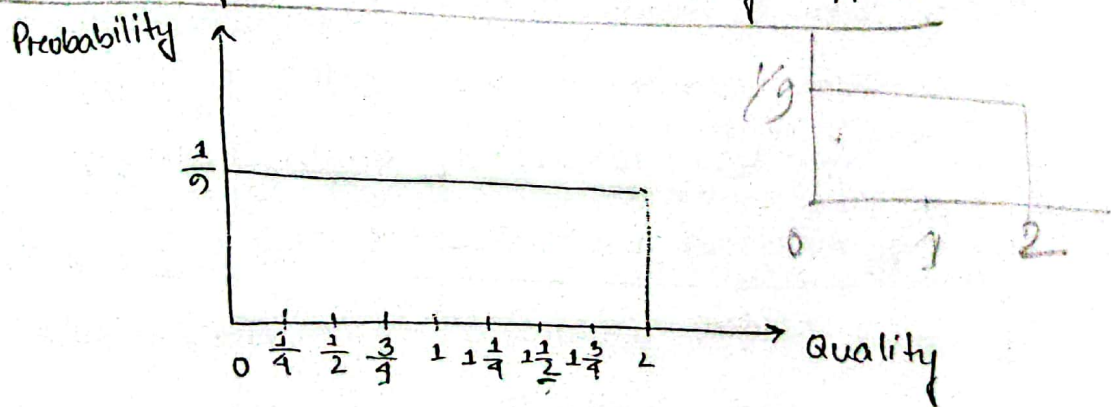


Fig: The availability of products of different quality.

Suppose the auctioneer calls out an initial price of $\$2,000$ per car. At this price, all owners know it is worthwhile to sell their cars, so all nine cars will be supplied. However,

non-owners, knowing only the distribution of quality but not the quality of each individual car, will make a best guess that a given car is of average quality; that is $Q=1$. They would not buy any cars at a price of \$2,000 because they are willing to pay only \$1,500 per unit of quality. They guess that all cars have a quality of 1 for a product of $\$1,500 \times 1 = \$1,500$, which is less than the \$2,000 asked. They would be willing to buy cars only if the price were less than or equal to \$1,500.

So, the auctioneer, perhaps trying to accommodate the potential buyers, tries a lower price, say \$1,500. Unfortunately, at this price, the owners of the two best cars will withdraw from the market. The owner of the car with two units of quality is receiving only $\$1,500 \div 2 = \750 per unit of quality; the owner of the car with $1\frac{3}{4}$ units will do the same way. The withdrawal of the two best cars causes the average quality of the seven remaining cars to fall. With nine cars, the average of the distribution was $Q=1$. Now at a price of \$1,500 per car, the best car offered will have a quality level of $\frac{3}{2}$ and the average quality will be $\frac{3}{4}$. Potential buyers would now be willing to pay only \$1,500 per unit of quality $\times \frac{3}{4}$.

unit of quality per average car, or \$1,125 for any car. Just as the previous price of \$2,000 per car was too high, the new price of \$1,500 is too high for buyers.

Surprisingly, in this example, no equilibrium that satisfies both buyers and sellers will be found. Akerlof saw this problem this way: when potential buyers know only the average quality of used cars, then market prices will tend to be lower than the true value of the top quality cars. Owners of the top quality cars will tend to withhold their cars from sale. In a sense, the good cars are driven out of the market by the lemons. Under what has become known as the lemons principle, the bad drives out the good until, as in some cases such as this one, no market is left.

Application of the lemons principle : Health Insurance

The application of the lemons principle to health insurance can be seen directly. In the following figure, let the horizontal axis measure the expected health expenditure levels of a population of n potentially insured people. Assume that their expected health expenditure levels for the insured ~~period~~ period range from a low of \$0 up to an expenditure level of $\$M$. The vertical axis represents the probability with a uniform distribution. The insurer must at least break even, which means that the premiums received from each insured must cover the insured population's average expenditure and other expenses.

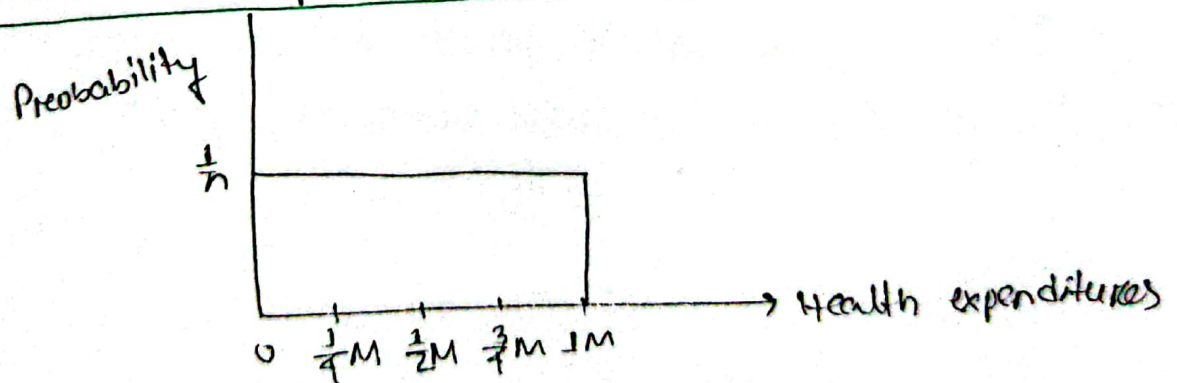


Fig: Uniform probability of expenditure.

This time the auctioneer attempts a first trial price of \$0! All potential beneficiaries would certainly demand

coverage at this price. Just as certainly, the insurance company expecting an average expenditure of $\$ \frac{1}{2}M$ would require a premium of at least $\$ \frac{1}{2}M$.

Following, Arrow's analysis, suppose the auctioneer tries a higher price, say $\$ \frac{1}{2}M$ hoping that this will clear the market. In this case, all potential beneficiaries who expect an expenditure level below $\$ \frac{1}{2}M$ will choose to self insure, that is leave the insurance market altogether, because this premium is higher than their privately known levels of health expenditure. When these healthier people leave the market, the average expected expenditures from $\$ \frac{1}{2}M$ to $\$ M$, rises to $\$ \frac{3}{4}M$. Thus the higher health risks tend to drive out the lower ~~wealthier~~ health risk people, and a functioning market may even fail to appear at all for some otherwise insurable health care risks.

Adverse selection refers to a situation where sellers have information that buyers do not, or vice versa. Adverse selection leads to economic inefficiencies.

Few people exactly know their future level of expenditures. Risk is the main reason for insurance. However, if the lower risks are grouped with higher risks and all pay the same premium, the lower risks face an unfavorable rate and will tend to underinsure. They sustain a welfare loss. Conversely, the higher risks will face a favorable premium and therefore overinsure; that is they will insure against risks that they would not otherwise insure against. This too, is inefficient. In addition to inefficiency, income will be redistributed from consumers who are lower risks to those who are higher risks.

inefficiency,
income redistribution

Group insurance can be a more useful mechanism to reduce adverse selection. Group plans enable insurers to implement experience rating, a practice where premiums are based on the past experience of the group or other risk rating systems to

group insurance

project expenditures.

Although experience rating can reduce adverse selection, it has come under increasing attack with the rapid growth of managed care plans such as health maintenance organizations (HMOs). HMOs receive a predetermined premium to provide the contracted health care for enrollees.

The Agency relationship. 3rd party.

An agency relationship is formed whenever a principal (for example, a patient) delegates ~~authority~~ ^{authority} decision making authority to another party, the agent. In the physician - patient relationship, the patient (principal) delegates authority to the physician (agent), who in many cases also will be the provider of the recommended services. The motive behind this delegation of authority is that the principals recognize that they are relatively uninformed about the most appropriate decisions to be made and that the deficiency is best resolved by having an informed agent. Thus asymmetric information and agency

are closely related phenomenon.

The perfect agent physician is one who chooses as the patients themselves would choose if only the patients possessed the information that the physician does. The patient's own interest focuses on his or her health. When any conflict arises, the perfect agent focuses on the patient's preferences, not his or her own. The problem for the principal is to determine and ensure that the agent is acting in the principal's best interests. Unfortunately the interests may diverge.

CEA sometimes provides a more practical approach to decision making than CBA. CEA compares the costs of achieving a particular nonmonetary objective. In cost effectiveness analysis, one assumes that the objective is desirable even if the benefits have not been evaluated in monetary terms. Though the valuation of benefits is avoided, the problems of determining costs remain.

The proper comparison of costs per output in CEA is based on the ratio of incremental costs to incremental output. Let the change in social costs incurred due to a particular project be $C_1 - C_0$ and let the gain in health output be $E_1 - E_0$. Then the various projects are compared by the ratio.

$$\text{CEA ratio} = \frac{C_1 - C_0}{E_1 - E_0}$$

The QALYs is a weighting system that assigns a value q , ranging from 1 (perfect health) to 0 (death) to represent quality of life for each year.

In its welfare economic version, the weights for QALYs are in principle derived by eliciting the individuals' preferences for different states of health. In practice,

the QALYs are often summed over groups of people.

However, once the weights are agreed upon and the QALYs are acceptably calibrated, then the measure becomes.

$$QALY = \sum_{i=1}^{n \text{ duration}} F_i q_i / (1+d)^i$$

where, F_i = probability that the person is still alive at age i .

d = time discount factor

q_i = quality weight between 0 and 1 assigned to each year of the person's remaining life.

Q Consider a health care project that costs \$10,000 today. It will provide benefits of \$4000 at the end of year 1, \$3500 at the end of year 2 and \$3000 at the end of year 3. If the discount rate is 6%, will this health care project be approved using CBA? Could your answers change if the discount rate is 0.5% (10.9%)?

Ans:

(Given)

the cost of health care project, $C_0 = \$10,000$

Benefit end of 1 year (B_1) = \$4,000

" " " 2 " (B_2) = \$3,500

" " " 3 " (B_3) = \$3,000

The discount rate, $d = 6\% = 0.06$

$$\text{Net present benefit, NPB} = \frac{B_0 - C_0}{(1+d)^0} + \frac{B_1 - C_1}{(1+d)^1} + \frac{B_2 - C_2}{(1+d)^2} + \frac{B_3 - C_3}{(1+d)^3}$$

$$= \frac{0 - 10,000}{(1+0.06)^0} + \frac{4,000 - 0}{(1+0.06)^1} + \frac{3,500 - 0}{(1+0.06)^2} + \frac{3,000 - 0}{(1+0.06)^3}$$

$$= -10,000 + \frac{4,000}{1.06} + \frac{3,500}{1.12} + \frac{3,000}{1.19}$$

$$= -10,000 + 3773.58 + 3125 + 2521.18$$

$$= -160.24$$